Problem sheet 6 for Rigid Analytic Geometry Summer Term 2026

From the lecture, recall that the G_+-space is a T_0-space X with a Grothendieck topology on $\mathfrak{O}_X$ such that $U = \bigcup_{V \in \mathcal{S}} V$ when $\mathcal{S} \models U$ and such that the empty sieve covers the empty set. A subset $\mathfrak{B} \subseteq \mathfrak{O}_X$ is called a G_+-base if it is an ordinary topology base and for all $U \in \mathfrak{O}_X$, the sieve generated by $\mathfrak{B}_U = \mathfrak{B} \cap \mathfrak{O}_U$ covers U.

Problem 1 (4 points)**.** *Let X be a T_0-space and $\mathfrak{B}$ a topology base for X such that the assumptions of Problem 9 hold. Show that X becomes a G_+-space when equipped with the Grothendick topology forcing the elements of $\mathfrak{B}$ to be quasicompact, and $\mathfrak{B}$ is a G_+-base of the G_+-space X!*

Problem 2 (4 points)**.** *Let X be a G_+-space and $\mathfrak{B}$ a G_+-base for X such that $\Omega, \Theta \in \mathfrak{B}$ implies $\Omega \cap \Theta \in \mathfrak{B}$. Let $X^*_{\mathfrak{B}}$ be the set of all $\xi_{\mathfrak{B}} \subseteq \mathfrak{B}$ such that the following conditions hold:*

- **V1:** *$X \in \xi_{\mathfrak{B}}$*
- **V2:** *If $\Omega, \Theta \in \xi_{\mathfrak{B}}$ then $\Omega \cap \Theta \in \xi_{\mathfrak{B}}$.*
- **V3:** *If $\Omega \in \xi_{\mathfrak{B}}$ and $\Omega \subseteq \Theta$ then $\Theta \in \xi_{\mathfrak{B}}$.*
- **V4:** *If $\Omega \in \xi_{\mathfrak{B}}$ and $\mathcal{S} \subseteq \mathfrak{O}_\Omega$ is a covering sieve for Ω then $\mathcal{S} \cap \xi_{\mathfrak{B}} \neq \emptyset$.*

In particular, $X^*_{\mathfrak{O}_X} = X^*$ is the set of van der Put points. Show that we have a bijection

$$\begin{aligned} X^* &\to X^*_{\mathfrak{B}} \\ \xi &\to \xi_{\mathfrak{B}} = \xi \cap \mathfrak{B} \end{aligned}$$

with the inverse map sending $\xi_{\mathfrak{B}} \in X^*_{\mathfrak{B}}$ to the set ξ of open subsets $U \subseteq X$ containing an element of $\xi_{\mathfrak{B}}$!

Problem 3 (3 points)**.** *Let X and $\mathfrak{B}$ are as in the first problem. Show that in the previous problem, condition V4 can be replaced by the following condition:*

> *If $\Omega \in \xi_{\mathfrak{B}}$, $n \in \mathbb{N}$ and $(\Theta_i)_{i=1}^n \in \mathfrak{B}^n$ are such that $\Omega = \bigcup_{i=1}^n \Theta_i$, then there is $i \in \mathbb{N}$ such that $1 \leq i \leq n$ and $\Theta_i \in \xi_{\mathfrak{B}}$.*

Let K be a field which is complete with respect to the non-discrete ultra-metric absolute value $|\cdot|$, and let $\mathbb{T}_1 = \mathbb{T}_{1,K}$. In the next few sheets we will investigate van der Put point of $X = \mathrm{Sp}\mathbb{T}_1$, equipped with the Grothendieck topology forcing the elements of $\mathfrak{Rat}_{\mathbb{T}_1}$ to be quasi-compact, establishing a bijection between the set of van der Put point and certain equivalence classes of continuous valuations of $\mathbb{T}_1$.

The following is a generalization of part of Problem 2 on problem sheet 1.

Problem 4 (3 points)**.** *Let $R \in (0,1]_{\mathbb{R}}$ and put*

$$N_R(f) = \max\bigl\{R^j\,|f_j| \;\big|\; j \in \mathbb{N}\bigr\}$$

for $f = \sum_{j=0}^{\infty} \in \mathbb{T}_1$. Then N_R is a multiplicative norm on $\mathbb{T}_1$.

Remark 1.

- *Thus, if $(0,\infty)_{\mathbb{R}}$ is equipped with its usual ordering and multiplicative group structure then*

$$\mathbb{T}_1 \xrightarrow{N_R} [0,\infty)_{\mathbb{R}} = \{0\} \cup (0,\infty)_{\mathbb{R}}$$

  is a valuation.
- *Of course $\mathbb{T}_1$ will only be complete with respect to N_R if $R = 1$, in which case $N_1(f) = \|f\,|\mathbb{T}_1\|$.*
- *If $R \in |K^\times|$, there is $k \in K^\times$ such that $R = |k|$ and N_R is simply the composition of $\|\cdot\,|\mathbb{T}_1\|$ with the K-algebra homomorphism*

$$\mathbb{T}_1 \to \mathbb{T}_1$$

$$f = \sum_{j=0}^{\infty} f_j T^j \to f(kT) = \sum_{j=0}^{\infty} k^j f_j T^j$$

  and the claim simply follows from Problem 2 on sheet 1. But we are interested in the general case where no such k exists.

If K is algebraically closed then the group $(0,\infty)_{\mathbb{R}} \,/\, |K^\times|$ is torsion free. As a consequence, if $R \in (0,1)_{\mathbb{R}} \setminus K^\times$ then for every $\nu \in |K^\times| \cdot R^{\mathbb{Z}}$ there is a unique $j_\nu \in \mathbb{Z}$ such that $\nu \in |K^\times|\, R^{j_\nu}$. In particular this can be applied to $\nu = N_R(f)$ when $f \in \mathbb{T}_1 \setminus \{0\}$.

Problem 5 (4 points)**.** *Let K be algebraically closed, $R \in (0,1)_R \backslash |K^\times|$, $f \in \mathbb{T}_1 \setminus \{0\}$, $\nu = N_R(f)$ and $j = j_\nu$. Show that there are $r_- \in (0,R)_{\mathbb{R}}$ and $r_+ \in (R,1)_{\mathbb{R}}$ such that*

$$|f(z)| = \nu\Bigl(\frac{|z|}{R}\Bigr)^j$$

holds for all $z \in K$ with $r_- \le |z| \le r_+$.

Let $B_{\le r}(0) = \bigl\{z \in K \;\big|\; |z| \le r\bigr\}$.

Problem 6 (6 points)**.** *Let K be algebraically closed. We identify K° with $\mathrm{Sp}\mathbb{T}_1$, identifying $z \in K^\circ$ with*

$$\mathfrak{m}_z = \bigl\{f \in \mathbb{T}_1 \;\big|\; f(z) = 0\bigr\}.$$

Let $R \in (0,1)_{\mathbb{R}} \setminus |K^\times|$, $m \in \mathbb{N}$ *and* $(f_j)_{j=0}^m \in \mathbb{T}_1^{m+1}$ *such that*

$$\langle f_0, \dots, f_n \rangle_{\mathbb{T}_1} = \mathbb{T}_1.$$

We consider the rational open subset

$$\Omega = \mathcal{R}(f_0 | f_1, \dots, f_m) = \big\{ z \in K^\circ \;\big|\; |f_j(z)| \leq |f_0(z)| \text{ for } 1 \leq j \leq m \big\}.$$

- *If* $N_R(f_j) \leq N_R(f_0)$ *holds for* $1 \leq j \leq m$*, show that there are* $r_- \in (0,R)_{\mathbb{R}}$ *and* $r_+ \in (R,1)_{\mathbb{R}}$ *such that* $B_{r_+}(0) \subseteq \Omega \cup B_{r_-}(0)$.
- *If there is a positive integer* $j \leq m$ *such that* $N_R(f_j) > N_R(f_0)$*, show that there are* $r_- \in (0,R)_{\mathbb{R}}$ *and* $r_+ \in (R,1)_{\mathbb{R}}$ *such that* $\Omega \cap B_{r+}(0) = B_{r_-}(0)$.

Note that the set of all Ω for which the first alternative occurs can be seen to satisfy V1–V3 from Problem 2 as well as the condition from Problem 3. It thus defines a van der Put point of $\mathrm{Sp}\mathbb{T}_1$. On the next few sheets we are going to extend these considerations, leading to a complete description of all van der Put points of $\mathrm{Sp}\mathbb{T}_1$ at least when K is algebraically closed.

Of the 22 points from this sheet, two are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday June 8. Good luck and nice holidays!